

# MBFD MID-MOUNT LADDER WORK GROUP

DAY 1 CONFIGURATION // INITIALIZATION

PROJECT CLASSIFIED // ASSET  
INITIALIZATION

JOB #41559AD

VEHICLE CLASS: PIERCE VELOCITY  
100' ASCENDANT AERIAL TOWER

MISSION PROFILE: TOOL SELECTION  
& STOWAGE ARCHITECTURE

STATUS: WORK GROUP

SESSION 1 - ACTIVE ■





## COMMAND DIRECTIVES

// **CONSENSUS IS KING:**  
Configuration decisions driven  
by Work Group vote.

// **THE OBJECTIVE:**  **HIGH PRIORITY**  
Evaluate, rank, and select  
loose equipment.

// **CRITERIA:**  
Cost,  
Interoperability,  
Serviceability,  
Lifecycle.



// **DELIVERABLE:**  
Finalized 'Top 3' list  
for procurement.



**HEALTH & SAFETY COMMITTEE**  
**FINAL PROCUREMENT AUTHORIZATION**  
AUTHORITY LEVEL: FINAL // STATUS: PENDING REVIEW  
**AWAITING INPUT**



**WORK GROUP**  
**TACTICAL COMMAND / DECISION MAKER**  
ACTIVE UNITS: 12  
DECISION CYCLE: IN PROGRESS **89% COMPLETE**



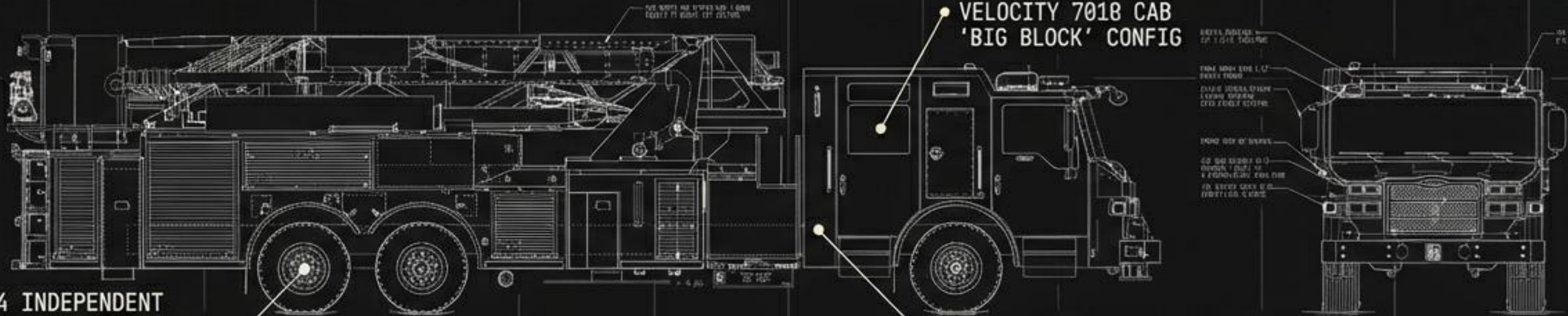
**SUPPORT SERVICES**  
**LOGISTICS & INTEL**  
STATUS: DEPLOYED // BES. 889.





## VELOCITY 100' ASCENDANT AERIAL TOWER

Inter



TAK-4 INDEPENDENT  
SUSPENSION // 24,000  
LB RATING //  
LOW CENTER OF GRAVITY

VELOCITY 7018 CAB  
'BIG BLOCK' CONFIG

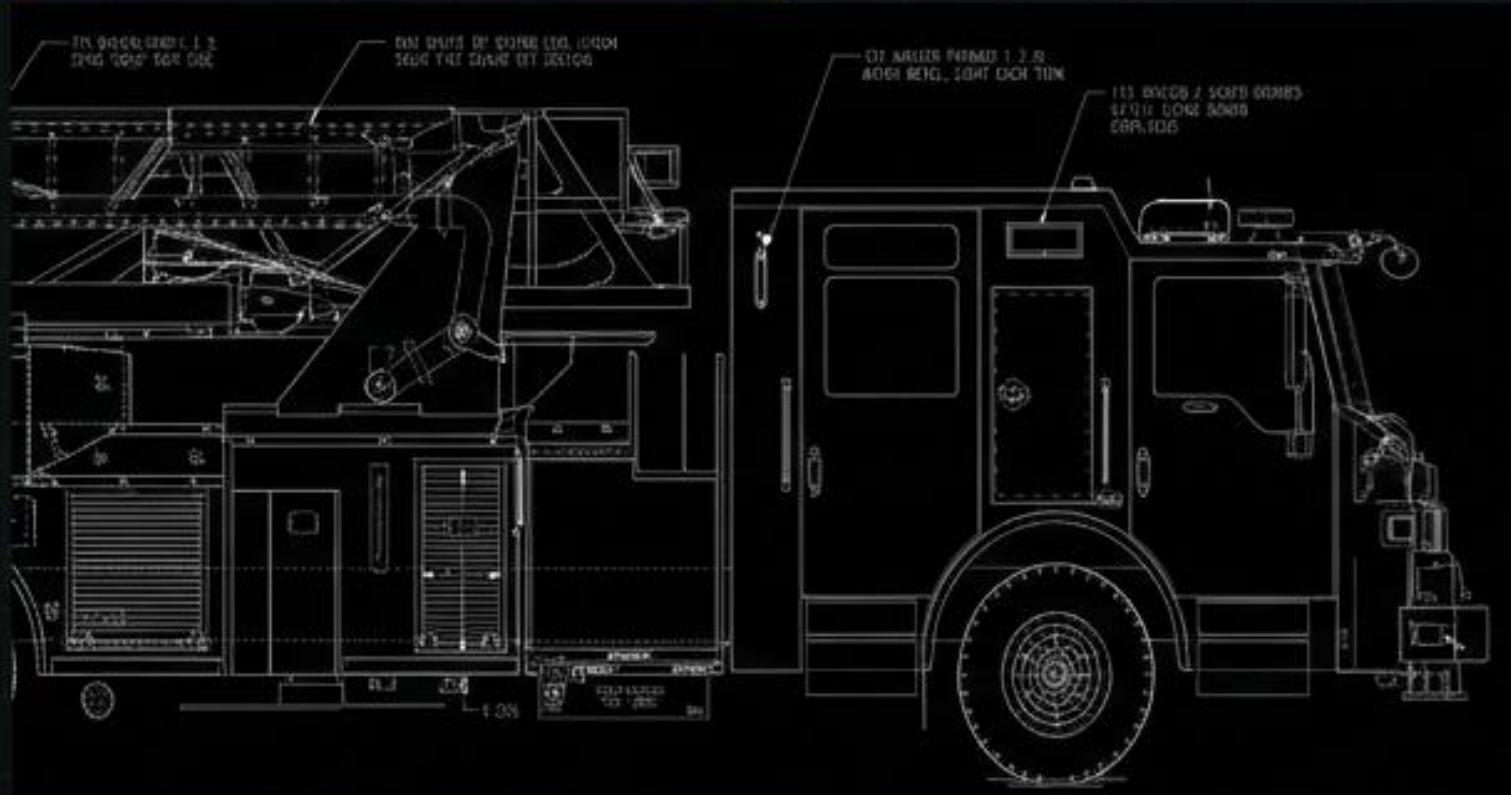
2,000 GPM WATEROUS PUMP (ESU1)

LENGTH: 528.50 INCHES (44' - 0.50")



# THE COMPARTMENT CHALLENGE: ASYMMETRICAL WARFARE

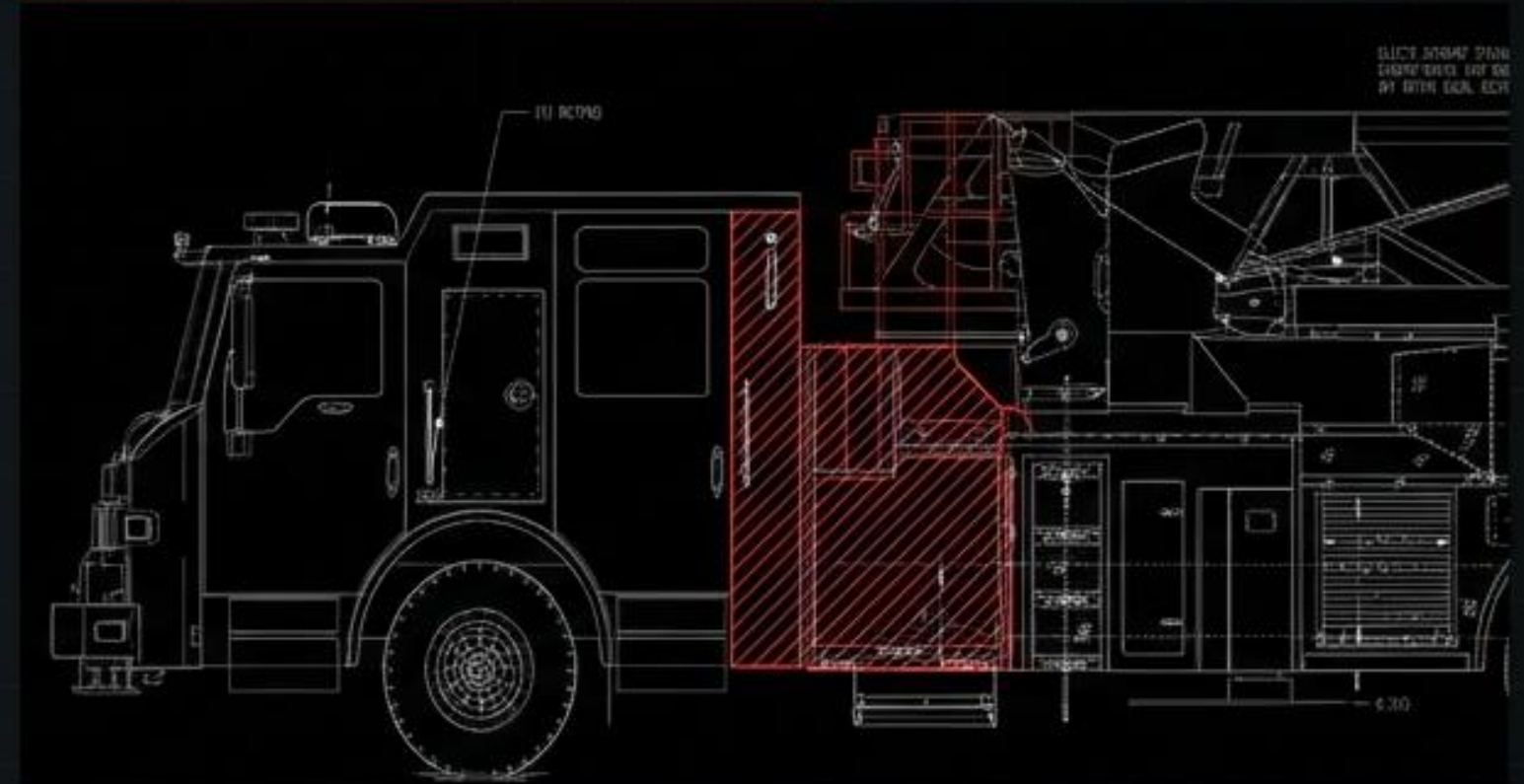
## DRIVER SIDE (L ZONES)



**FULL HEIGHT & FULL DEPTH.**

Primary Bulk Storage (Fans, Saws, Rescue Kits).

## OFFICER SIDE (R ZONES)



**FULL HEIGHT / REDUCED DEPTH.**

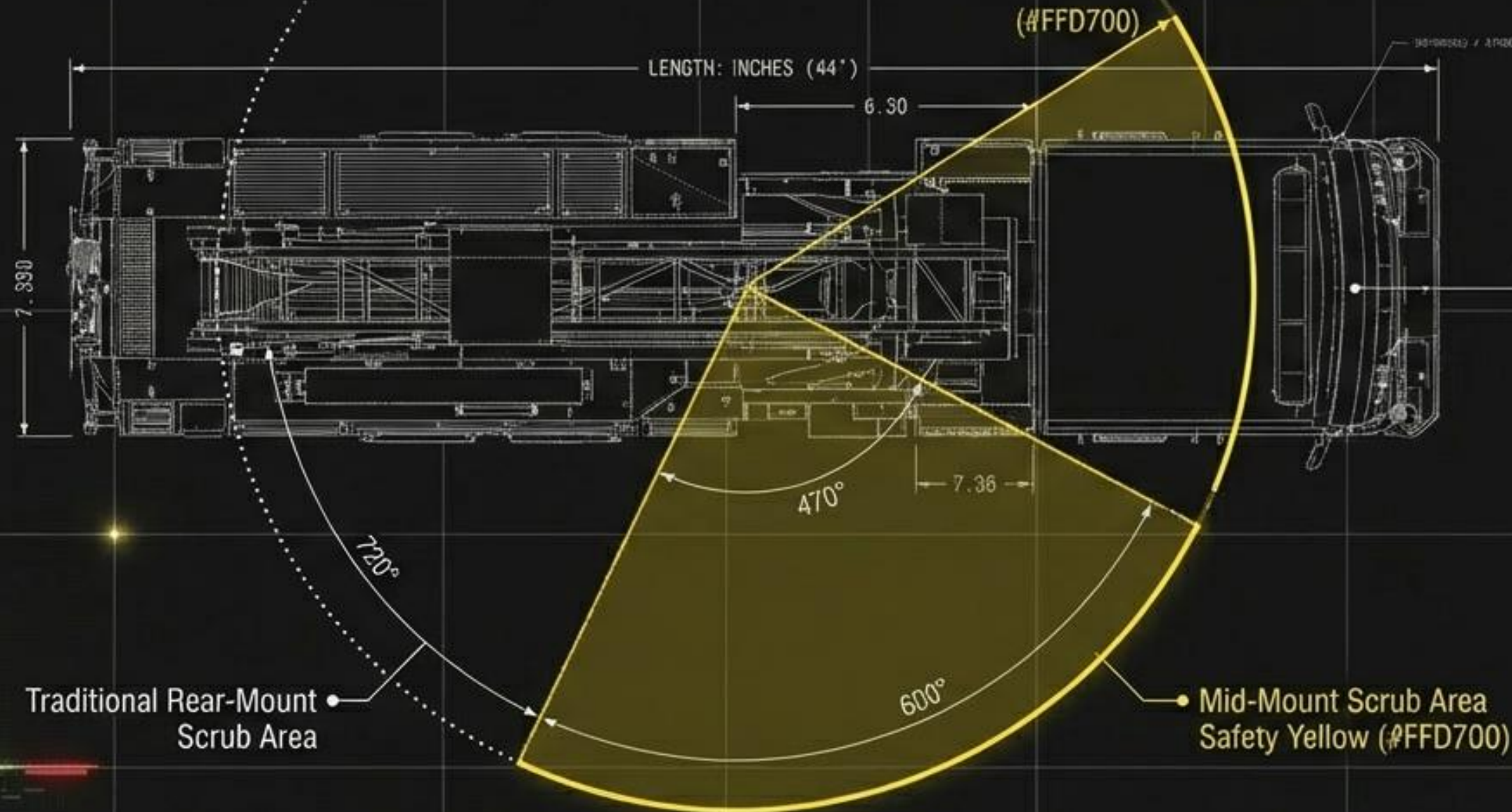
## Torque Box Intrusion.

**STRATEGIC IMPLICATION: NO COPY/PASTE.  
WE MUST OPTIMIZE FOR NARROW-PROFILE TOOLS.**





# OPERATIONAL PARADIGM SHIFT



## SYSTEM ALERT

### DEPLOYMENT GEOMETRY

Mid-mount turntable placement changes spotting distances.

## SYSTEM ALERT

### TRAINING IMPERATIVE

"Fend-off" positioning & cab clearance protocols.

## SYSTEM ALERT

### SWING-OUT FACTOR

Tail-swing minimized (2 ft) / Cab obstruction increased.



# FIXED ASSETS VS. OPEN ARCHITECTURE

## LOCKED (FIXED ASSETS)



SCENE LIGHTING



CHARGING INFRASTRUCTURE

## UNLOCKED (CLEAN SLATE)



- EXTRICATION?
- FORCIBLE ENTRY?
- SAWS?



### DECISION POINT

### STRATEGIC QUESTION

**DECISION POINT: DO WE COMMIT EXCLUSIVELY TO THE DEWALT PLATFORM TO MATCH THE TRUCK ECOSYSTEM?**



# FLUID DYNAMICS: FEEDING THE 2,000 GPM PUMP



HEAVY HYDRANT  
(5" STORZ)



OASIS VALVE /  
JUMBO SIAMESE



2,000 GPM PUMP



1,500 GPM AERIAL FLOW



HANDLINES



OBJECTIVE

Uninterrupted high flow  
water supply



# MARKET INTELLIGENCE: DHS SAVER REPORT

| MANUFACTURER                         | WEIGHT (LBS) | NFPA RATING | IP RATING |
|--------------------------------------|--------------|-------------|-----------|
| Amkus (ION)                          | 58.4         | Yes         | IP68      |
| Holmatro (Pentheon)                  | 33.7         | Yes         | IP68      |
| Hurst (Jaws of Life) (eDRAULIC EWXT) | 46.6         | Yes         | IP67      |
| TNT (Storm E-Tools)                  | 56.8         | Yes         | IP68      |
| Genesis (EFORCE 3.0)                 | 48.7         | Yes         | IP67      |
| ResQtec (EDD/P-line)                 | 49.6         | Yes         | IP55      |

DATA INSIGHTS

// WEIGHT VARIANCE:  
33.7 lbs - 58.4 lbs.

// ERGONOMICS: Critical for  
mid-mount deployment.

// ENVIRONMENT: IP Ratings  
essential for saltwater  
protection.

// POWER: COTS vs.  
Proprietary Batteries.





# THE POWER SOURCE DEBATE

## COTS (COMMERCIAL OFF-THE-SHELF)

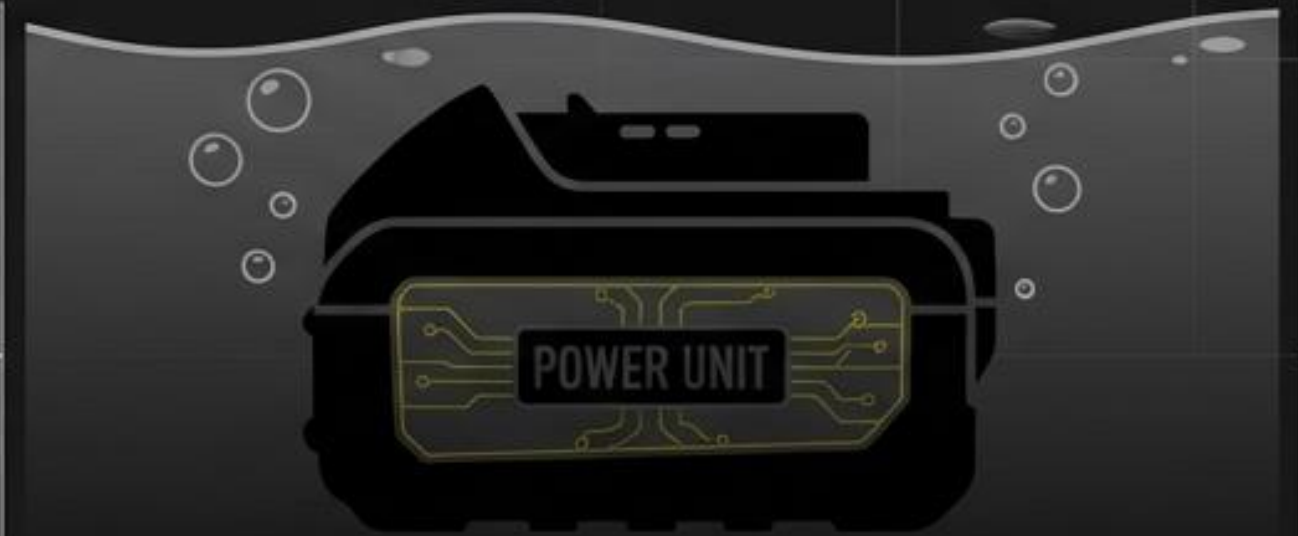


**PROS:** Available anywhere, cheap, cross-compatible (Drills, Saws).

**CONS:** Lower IP ratings.

**VS**

## PROPRIETARY PLATFORM



**PROS:** Engineered for the tool, Submersible (IP68).

**CONS:** Single-source procurement, expensive.

**DECISION MATRIX:**  
LOGISTICAL EASE VS. UNDERWATER CAPABILITY.





# CANDIDATE PROFILE: AMKUS RESCUE SYSTEMS



## TECHNICAL SPECIFICATION MODULE



- PLATFORM: DeWALT 60V FLEXVOLT (NON-PROPRIETARY)
- FEATURE: ALUMINUM HOUSING / COMPACT DESIGN
- ACCESSORY INTEL: GATOR TIPS & EXTENDED REACH TIPS (ERT)



TACTICAL ADVANTAGE: BATTERIES MATCH TRUCK INFRASTRUCTURE





# CANDIDATE PROFILE: TNT RESCUE 'STORM SURGE'



## TECHNICAL SPECIFICATION MODULE

- PLATFORM: MILWAUKEE M18 / M28 REDLITHIUM
- PERFORMANCE: HIGH NFPA CUT TESTS (A8/B9/C8/D9/E9)
- DURABILITY: IP54 RATING



- TACTICAL ADVANTAGE: COMMONALITY WITH EXISTING FIRE SERVICE TOOLS





# CANDIDATE PROFILE: HURST EDRAULIC



## TECHNICAL SPECIFICATION MODULE

-  PLATFORM: PROPRIETARY HIGH-PERFORMANCE BATTERY
-  THE 'SALTWATER' FACTOR: IP58/68 RATED OPTIONS
-  PERFORMANCE: 'UNSTOPPABLE' FORCE RATINGS



 TACTICAL ADVANTAGE: TRUE UNDERWATER CAPABILITY FOR CANAL/BAYSIDE INCIDENTS





# STABILIZATION & STRUCTURAL INTEGRITY

## STORAGE ARCHITECTURE: L1/R1 PASSTHROUGH

J5TSBA1N5 R040  
59E22:  
01016: 800R  
150670: 228CH  
52605:  
LEASTN: 8

WIDTH:  
80CM

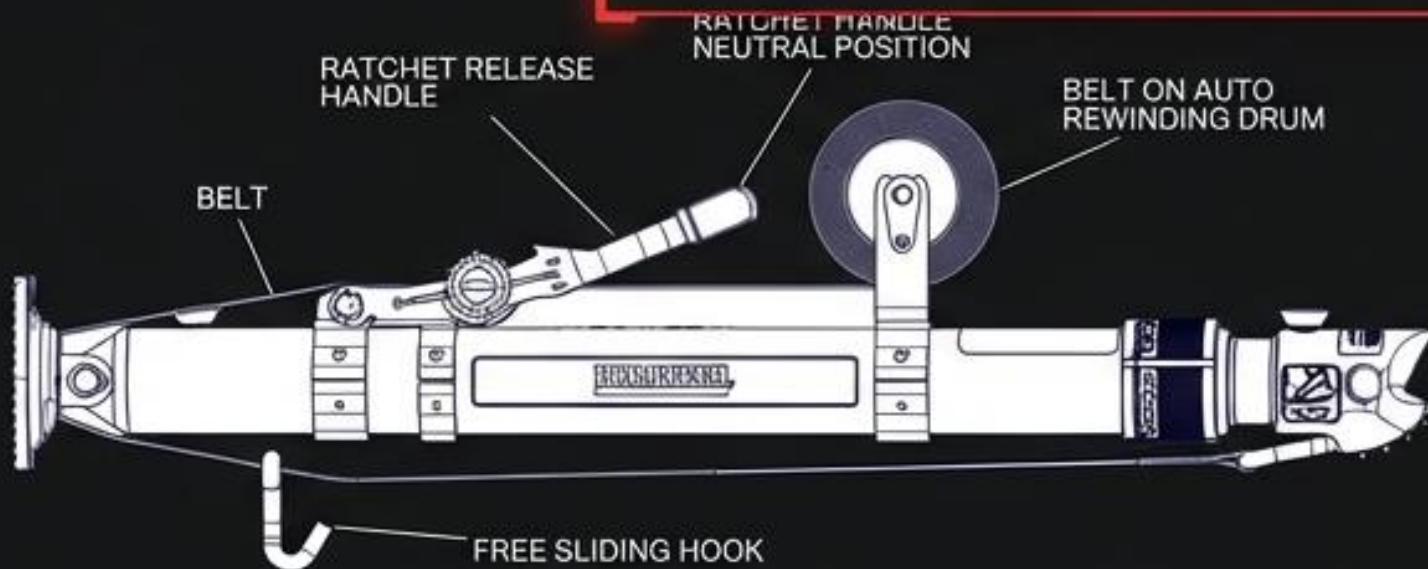


TRANSVERSE TUNNEL

LENGTH: 120CM



**REQUIREMENT: MUST HANDLE VEHICLE STABILIZATION  
& STRUCTURAL COLLAPSE SHORING.**



▶ PARATECH // GOLD STANDARD // TWIST-LOCK

**RES-Q-JACK™**  
STABILIZATION ELEVATED



▶ RES-Q-JACK // LIGHTWEIGHT ALUMINUM // X-STRUT





# VENTILATION: THE ZERO-EMISSION FRONTIER



**ZERO DIRECT EMISSIONS //**  
**SAFE FOR INDOOR /**  
**CONFINED SPACE.**

No exhaust fumes, allowing  
operation in enclosed  
environments without risk.



**INSTANT START //**  
**REDUCED MAINTENANCE.**

Electric motor delivers immediate  
power and requires significantly less  
upkeep than gas engines.



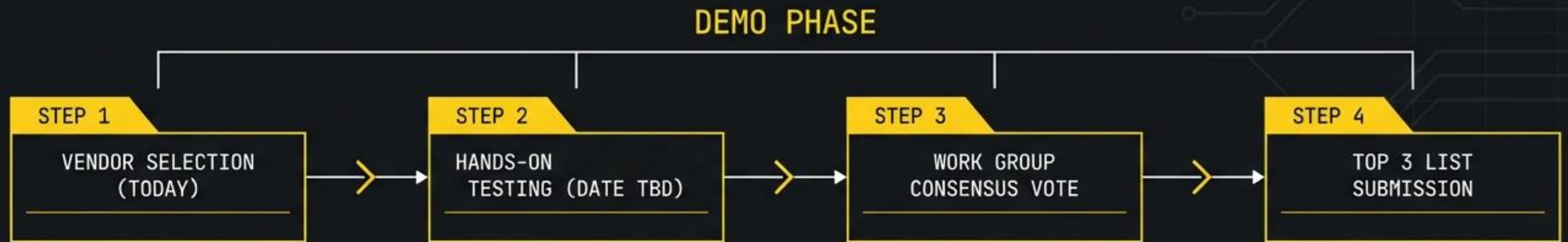
**STORAGE: REAR**  
**COMPARTMENT (L4) //**  
**INTEGRATED CHARGING**  
**REQUIRED.**

Dedicated storage space with  
built-in charging infrastructure  
is essential for operational  
readiness.

EVALUATION METRIC: RUNTIME VS. REINFORCED CONCRETE PERFORMANCE







**THE FLOOR IS YOURS.**

